

The knuth style aka YayT_EX

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1. Introduction

If you know anything about T_EX—and since you’re a Pandoc user, you probably do—you know that Computer Modern is the standard font. I have some enmity with Computer Modern, though—I think that, in the grand scheme of things, it is often a non-choice. I would be willing to bet that a vast majority of papers on the arXiv use Computer Modern as their main font, not because they particularly like it, or that it’s always the best tool for the job, but because it’s there from the start. Like the default font in Word or LibreOffice, you use Computer Modern because you don’t particularly care about what your document looks like, because you’re more worried about peer reviewers tearing your article to shreds than making the document look unique or pretty.

The thing is, I like Computer Modern despite its flaws. People make a big deal about how pretty L^AT_EX output is, and I can’t blame them—some particularly wonderful examples can found in *SIAM Review* articles from the 1980s or so—but it’s not always perfect. Once it got converted to PostScript, Computer Modern became *spindly* and difficult to read on screens. Thankfully, as of a few years ago, Antonis Tsolomitis began work on New Computer Modern—one of the greatest features he added is a new “book” weight which recaptures Computer Modern’s old beauty by giving it some heft.

The knuth style is dedicated to giving your documents a *learned* edge. Personally, there are times that I just want to use Computer Modern because it looks nice, and because it’s the default style. Likewise, this is a great style for heavy amounts of mathematical typesetting—New Computer Modern has loads of features that make mathematics render wonderfully.

Unfortunately, there aren’t many weights available in the sans font, so that keeps me from exercising my normal typesetting panache (or lack thereof).

2. Standard demo

The following text is taken from Lieb, E., Schultz, T., Mattis, D. (2004).¹
Please, go read the actual paper, it's great.

2.1. Formulation

The first model consists of N spin $1/2$'s (N even) arranged in a row and having only the nearest neighbor interactions. It is

$$H_\gamma = \sum_i [(1 + \gamma) S_i^x S_{i+1}^x + (1 - \gamma) S_i^y S_{i+1}^y],$$

where γ is a parameter characterizing the usual degree of anisotropy in the xy -plane. Because the Hamiltonian only involves the x - and y -components of the spin operators, we call this model the XY model.

To solve the XY model, we first introduce the raising and lowering operators

$$a_i^\dagger = S_i^x + i S_i^y \text{ and } a_i = S_i^x - i S_i^y$$

in terms of which the Pauli spin operators are

$$S_i^x = (a_i^\dagger + a_i)/2; \quad S_i^y = (a_i^\dagger - a_i)/2i; \quad S_i^z = a_i^\dagger a_i - \frac{1}{2}$$

and the Hamiltonian is

$$H_\gamma = \frac{1}{2} \sum_i [(a_i^\dagger a_{i+1} + \gamma a_i^\dagger a_{i+1}^\dagger) + \text{h.c.}].$$

The ground state Ψ_0 is the state with no elementary excitations:

$$\eta_k \Psi_0 = 0, \text{ all } k.$$

The ground-state energy, according to (A-12) is

$$E_0 = -\frac{1}{2} \sum_k \Lambda_k.$$

¹Two Soluble Models of an Antiferromagnetic Chain. In: Nachtergaele, B., Solovej, J.P., Yngvason, J. (eds) Condensed Matter Physics and Exactly Soluble Models. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-06390-3_35

In the limit $N \rightarrow \infty$, the sum can be replaced by an integral giving

$$\begin{aligned}\frac{E_0}{N} &= -\frac{1}{4\pi} \int_{-\pi}^{+\pi} [1 - (1 - \gamma^2) \sin^2 k]^{1/2} dk \\ &= -\frac{1}{\pi} \mathcal{E}(1 - \gamma^2),\end{aligned}$$

where $\mathcal{E}(k^2)$ is one of the complete elliptic integrals (13).